

Predicting Credit Spreads Using Macroeconomic Indicators

Objective

- The primary aim of this project was to explore the relationships between macroeconomic variables—such as the VIX, Treasury yields, and other indicators—and credit spreads, and to build a predictive model that could support decision-making on a credit trading desk.

Methodology

Data Audit

- We began by converting all date entries to a consistent datetime format and verified that the dataset followed a weekly frequency with no missing periods. Duplicate rows were checked (none found), and missing values were addressed through appropriate imputation. We also confirmed the dataset's date range and visually inspected the key variables—VIX, 13W Treasury Yield, 10Y Treasury Yield, and Credit Spread—to identify any structural breaks or anomalies. These steps ensured the dataset was clean, complete, and ready for time series modelling.

EDA & Macro Relationship Exploration

- This phase focused on identifying relationships between macroeconomic indicators and credit spreads to guide feature design and model structure. We began with correlation analysis across both the full dataset and volatility-based regimes. The 10Y Treasury yield showed a strong positive correlation with credit spreads, while the 13W T-Bill yield was strongly negative—both relationships held across regimes. VIX, however, had little contemporaneous correlation, suggesting limited linear predictive value. Lag analysis revealed that VIX exhibited mild positive correlation with credit spreads at lags of 5–10 days, indicating some delayed influence. In contrast, both Treasury yields had their peak effects at lag 0, consistent with contemporaneous macro-spread linkages. Segmenting by volatility regime highlighted structural shifts: the 10Y yield's relationship with spreads was even stronger during low-volatility periods, while VIX remained largely uninformative in linear form. These insights aligned with financial intuition—e.g., falling T-Bill yields during easing cycles often coincide with wider spreads. Overall, the EDA helped us identify key predictive signals, refine feature engineering, and remove deterministic or misleading variables to support robust modelling.

Feature Engineering

- We created over 60 new features from the original data to help the model better understand market patterns. These included past values, moving averages, volatility measures, and indicators of momentum or trend reversals. To avoid overfitting, we removed features that had a direct mathematical relationship with the target (like the 13W–10Y yield spread). Using model-driven importance and forward-looking correlations, we narrowed this down to 30 high-quality features. The most useful included the 10-day z-score of credit spreads (capturing how far spreads deviate from normal), the steepness of the yield curve, and momentum features showing recent spread direction. These helped the model detect signals without relying on overly obvious or circular inputs.

ML Model

- Our initial model looked perfect — it was predicting the direction of credit spreads with 100% accuracy. But that raised a red flag. After digging deeper, we found the reason: the model was relying on features that were deterministic — meaning they directly defined the credit spread, such as the difference between the 13-week Treasury yield and the 10-year yield (since $\text{credit spread} = 13W - 10Y$). Including features like these (or their moving average differences) caused the model to "cheat" by learning from the target variable itself. To fix this, we removed all features that were mechanically tied to the credit spread. Then, we focused on finding true leading indicators — features that tend to move before spreads change. We did this by checking the correlation of each variable with the 5-day forward return of credit spreads. Interestingly, many of the most predictive features had negative correlations, meaning when they rose, spreads often fell (and vice versa). Using those signals, we created new features that captured macro relationships more effectively — like changes in volatility, interaction terms, and momentum-style features. Finally, we trained models like Random Forest and XGBoost and used their built-in feature importance metrics to filter out the least useful inputs. The resulting model, an ensemble combining the best versions of RF and XGB, delivered strong performance and over 90% directional accuracy, using only sound, forward-looking signals.

Results

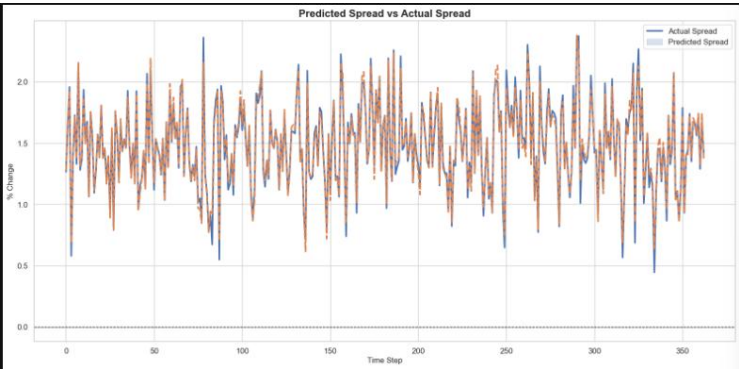
The final ensemble model (Random Forest + XGBoost) delivered strong and robust performance:

- Test RMSE: **0.0586**
- Test MAE: **0.0423**
- Directional Accuracy: **97.8%**

The model consistently predicted the direction of spread movement, a critical metric for trading. XGBoost individually outperformed in nested cross-validation (**RMSE: 0.0554 ± 0.0126**), confirming its generalization. Early versions showed artificially high accuracy due to deterministic inputs. After removing those and re-engineering features, performance stabilized, and interpretability improved. Visuals of predicted vs. actual spreads confirmed tight tracking and well-calibrated confidence intervals. Percentage change analysis further validated the model’s responsiveness across volatile and stable regimes. In summary, the final model achieves strong out-of-sample predictive performance, with a high signal-to-noise feature set grounded in both statistical evidence and macro theory

Interpretation & Use Case for Credit Trading Desk

The model can serve as a valuable tool for a credit trading desk by generating forward-looking predictions of credit spread movements across different time horizons. For daily use, next-day forecasts can help traders anticipate short-term spread widening or tightening, allowing them to adjust risk positions accordingly—for example, purchasing credit default swap (CDS) protection ahead of anticipated stress or reducing exposure in tightening environments. On a weekly horizon, the model supports tactical decisions such as rotating between investment-grade and high-yield bonds or shifting exposures between sectors based on evolving macroeconomic signals. For longer-term strategies, monthly forecasts can inform strategic portfolio adjustments, like altering duration or credit quality allocations as volatility regimes shift. By linking the model to a live macro data feed, the desk can produce automated, updated forecasts each morning before the market opens. These predictions can trigger trade signals when thresholds or confidence intervals are met—for instance, a predicted 10–15 bps spread widening with high certainty might prompt a shift into long-volatility positions or hedging corporate bond exposure in sectors with elevated risk. The model also flags potential mispricings where actual spreads deviate significantly from predicted values, offering mean-reversion or breakout trade opportunities. Over time, such tools help institutional desks systematize discretionary decision-making and improve overall risk-adjusted performance.



	Actual Spread	Predicted Spread	Difference	Error %
353	1.4400	1.4077	0.0323	-2.24
354	1.7389	1.7274	0.0115	-0.66
355	1.3506	1.3843	0.0336	2.49
356	1.6738	1.7062	0.0323	1.93
357	1.6672	1.6664	0.0008	-0.05
358	1.5620	1.5749	0.0129	0.82
359	1.7203	1.7417	0.0214	1.24
360	1.2890	1.3224	0.0333	2.59
361	1.6721	1.7382	0.0661	3.95
362	1.4170	1.3748	0.0422	-2.98

Limitations

- While the model demonstrates strong performance, particularly in directional accuracy, it has some limitations. It relies heavily on macroeconomic data and may underperform during periods driven by idiosyncratic or event-specific credit risk, such as sudden downgrades or geopolitical shocks. Its effectiveness also varies across regimes—features that work well in stable markets may lose signal in stressed conditions. Moreover, the model's static training approach means it may not adapt quickly to structural shifts in monetary policy or market sentiment. To address these challenges, a developed version of the model should integrate live data feeds, regular retraining with adaptive learning, and stress-testing capabilities. Enhancements such as incorporating sentiment data, scenario analysis, and trader overrides would help bridge macro-model output with real-time desk needs, making it a more robust tool in dynamic trading environments.